①

ELL 101

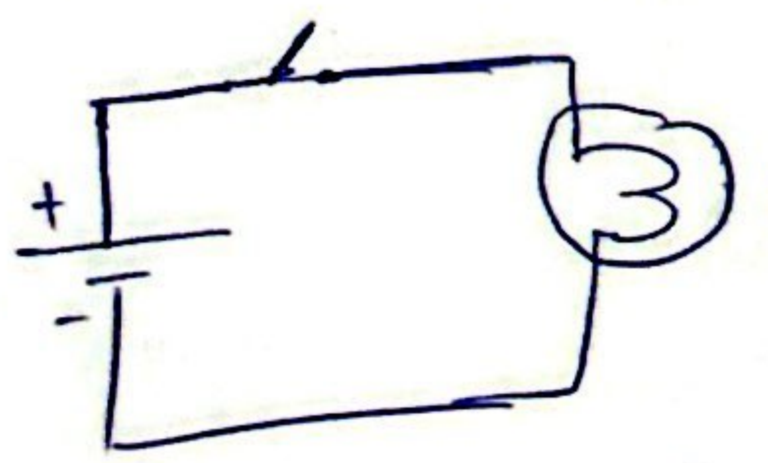
- What comes to mind when you think 'electrical engineering'?

- Name a device/gadget without which you can't live

- Name a technology without which you can't live

- All these devices/tech have circuits in them
  open a mobile charger & show

- In this course, we learn about these circuits
  ↓
  circulation such as blood in body ; water



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→ Electricity is everywhere around us : Atoms, molecules, nervous system

- like air of modern times

- sustainability ⟷ electrification
  [transportation, cooking, teaching]

- other components of elec eng.
  → control
  → communication
  → power
  → electronics
  →

Basic Concepts

1.) Charge (Q): most basic quant. in elec. circuit
elec. prop. of atomic particles measured in coulombs (C)

1 elect. $= -1.602 \times 10^{-19}$ C

2.) Current (I):
rate of change of charge, meas. in Amp.

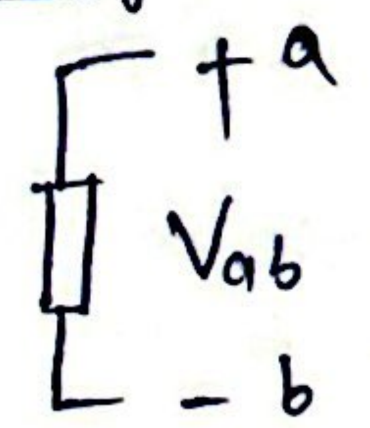
$$i = dq/dt$$

1 Amp = 1 coulomb/sec.

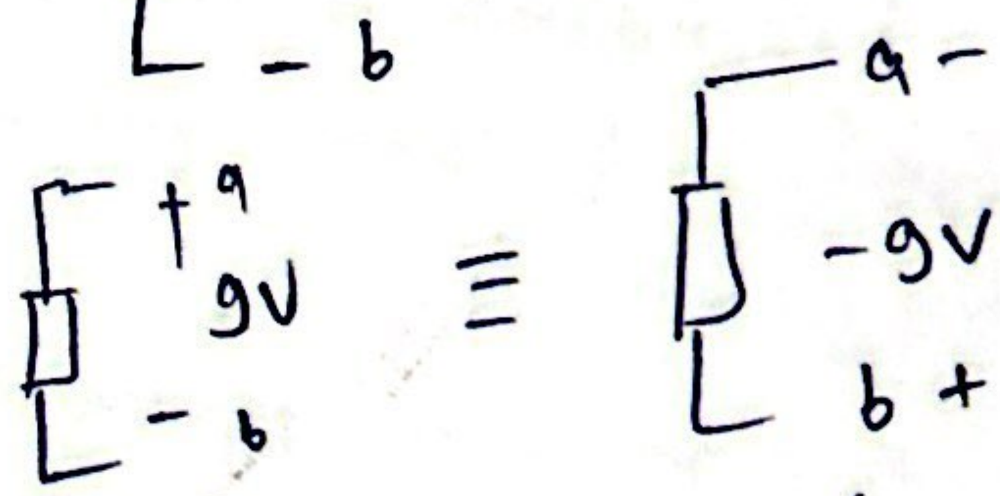
$$Q = \int_{t_0}^{t} i\,dt$$

DC & AC current

3.) Voltage



$V_{ab} = -V_{ba}$



$V_{ab} = 9V$ $\quad$ $V_{ba} = -9V$

Voltage (potential diff) is the energy required to move a unit charge from a ref. point (b) (-) to another point (+) (a) meas. in volts

$$V_{ab} = dw/dq$$

1 Volt = 1 J/coulomb

4.) Power : time rate of change in energy absorption or expansion (w)

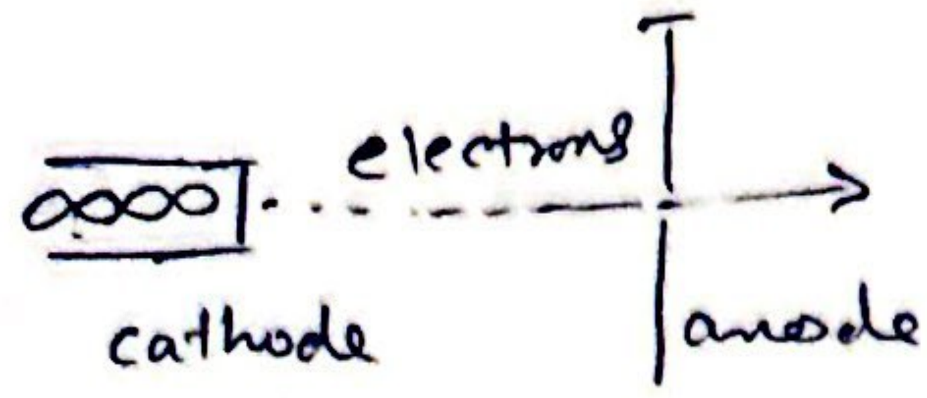
$$P = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v\,i$$

5.) Energy : Energy supplied or absorbed by an element from t_0 to t

$$w = \int_{t_0}^{t} P\,dt = \int_{t_0}^{t} v\,i\,dt$$

② 1 wh = 3600 Joules

Ex:



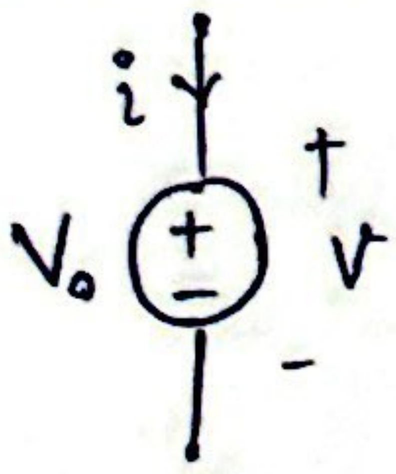
Calculate the power supplied to a beam of 50 million billion electrons per second through a potential diff of 20,000 Volts.

$$i = 1.6\times10^{-19} \times 50\times10^{6}\times10^{9} = 0.008A$$

$$P = v\,i = 2\times10^{4}\times 8\times10^{-3} = 160\,W$$

Basic Circuit Elements

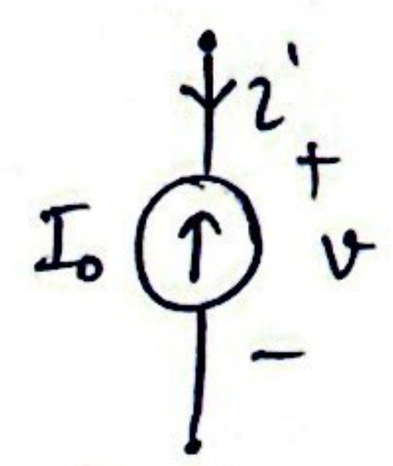
1.) Source



$v = V_0$

ideal voltage source

independent of i, or load connected

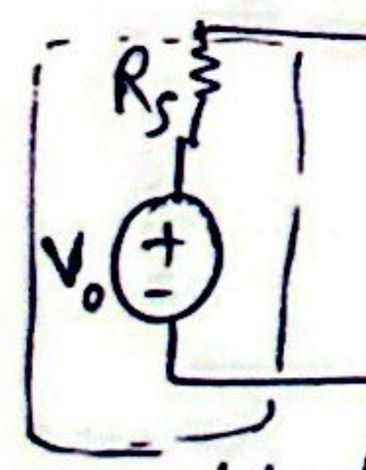


$i = -I_0$

ideal current source

independent of v or Load connected

⇒ but practically, no ideal Sources

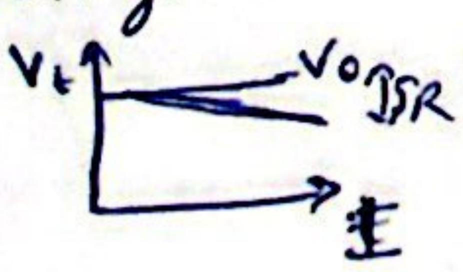


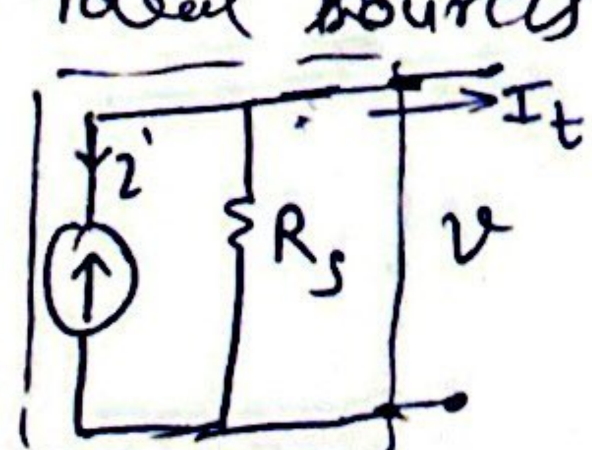
R_s: internal resistance

V_t

non-ideal voltage source

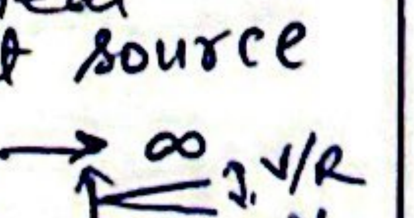
$R_s \to 0$





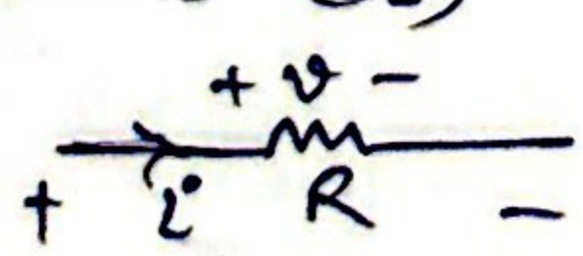
non-ideal current source

$R_s \to \infty$



2) Resistance

Resistance : R of an material denotes it's ability to resist the flow of electric current, measured in ohms (Ω)



$v = iR$

$R = \frac{\rho l}{A}$

A

show slide 24-25 (R of common materials) (Bhimsingh slides)

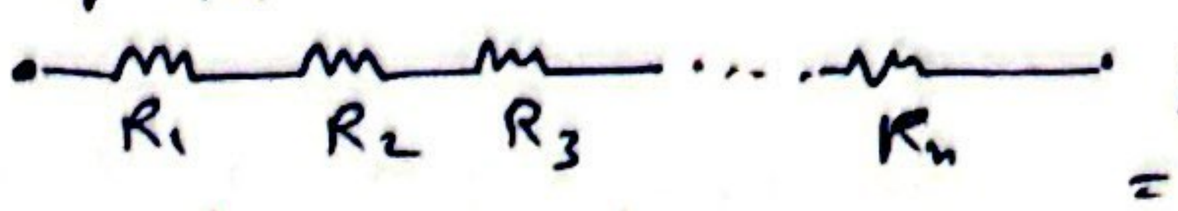
⇒ conductance : $G = \frac{1}{R}$ mho ℧ or Siemens (S)

⇒ Power dissipated by Resistor in heat

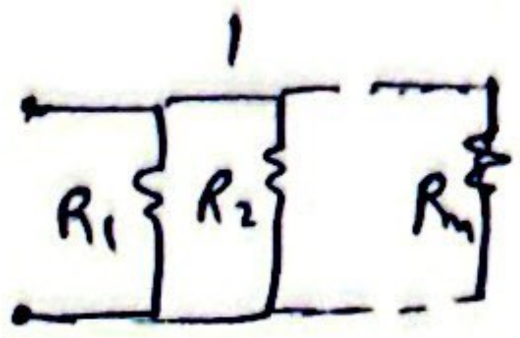
$P = v\,i = i^2R = v^2/R$

(irreversible)

⇒ Series & Parallel connection of R



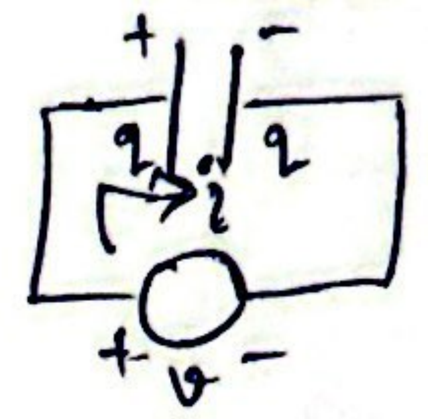
$R_{eq} = R_1 + \ldots R_n$



$\frac{1}{R_{eq.}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots \frac{1}{R_n}$

3.) Capacitor (C)

Capacitor (C) : two conducting plates, separated by an insulator



$$q = Cv$$

$$dq/dt = C\frac{dv}{dt}$$

$$i = C\,dv/dt$$

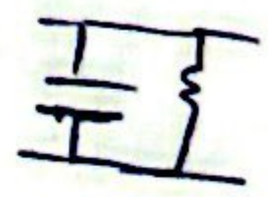
ability to store energy in stationary charges or electric field

⇒ Energy stored : $E = \int_{t_0}^{t} v\,i\,dt = C\int_{v(t_0)}^{v(t)} v\,dv$

$= \frac{1}{2}Cv^2$

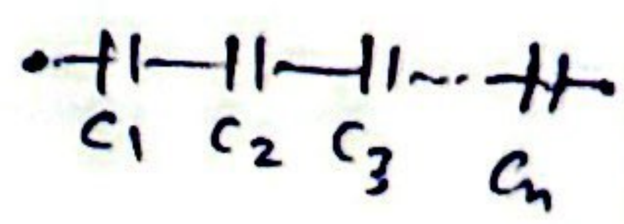
⇒ an abrupt change in voltage requires infinite current ⇒ Voltage can not change abruptly

⇒ non ideal capacitor



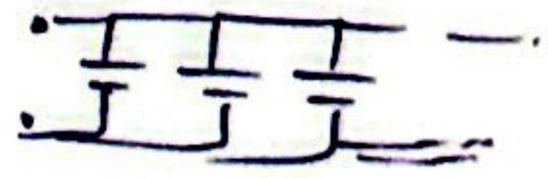
⇒ cap in series

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots \frac{1}{C_n}$

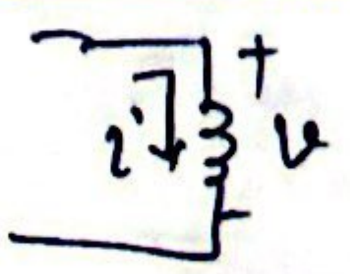


⇒ Cap in parallel

$C_{eq.} = C_1 + C_2 + \ldots$



4.) Inductor



$v = L\,di/dt$

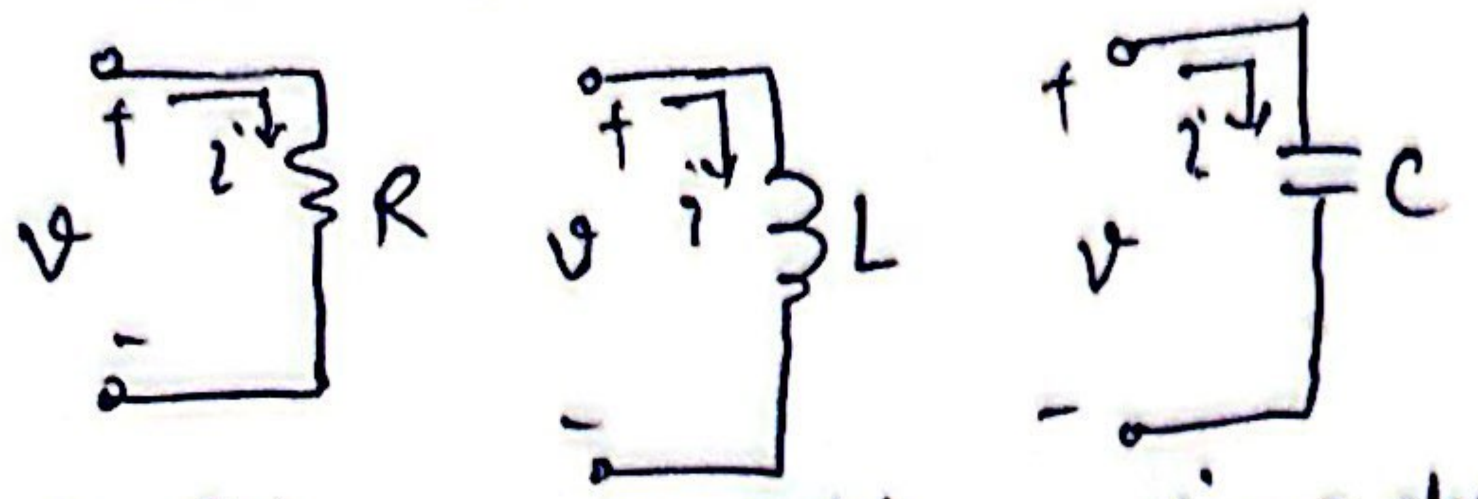
ability to store energy in moving charges or magnetic field

⇒ Energy stored : $E = \int_{t_0}^{t} v\,i\,dt = \frac{1}{2}Li^2$

⇒ an abrupt change in current requires ∞ voltage

⇒ series: $L_1 + L_2 + \ldots$, Parallel: $\frac{1}{L_{eq}} = \frac{1}{L_1} + \ldots$

③ Summary:

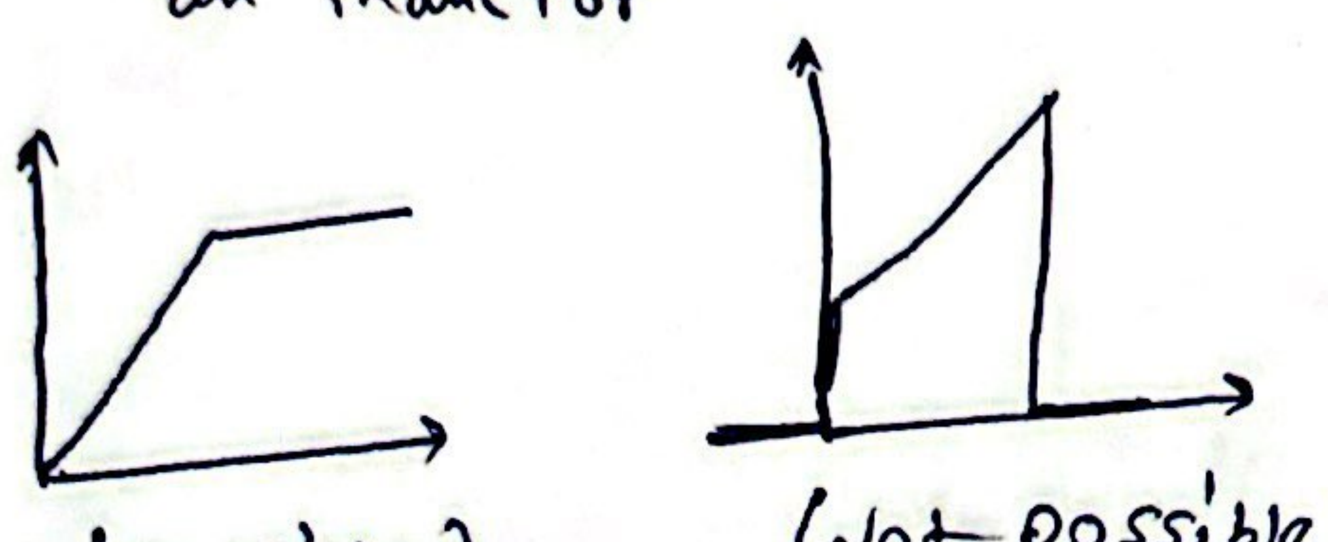


$v = Ri$ $\qquad$ $v = L\frac{di}{dt}$ $\qquad$ $i = C\frac{dv}{dt}$

$i = v/R$ $\qquad$ $i = \frac{1}{L}\int_{-\infty} v\,dt$ $\qquad$ $v = \frac{1}{C}\int_{-\infty} i\,dt$

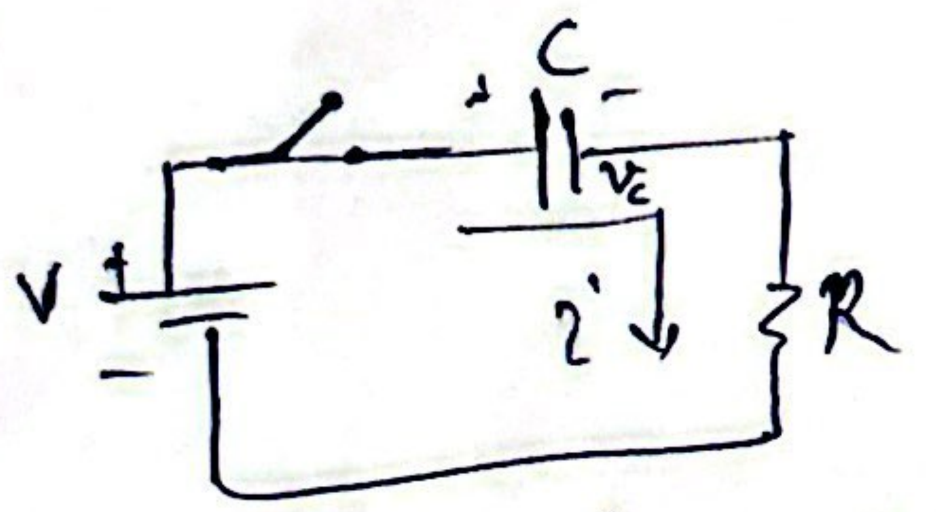
⇒ Voltage across capacitor & current in an inductor can not change instataneously.

Qu: which can be current through an inductor



(possible) $\qquad$ (Not possible)

Qu:

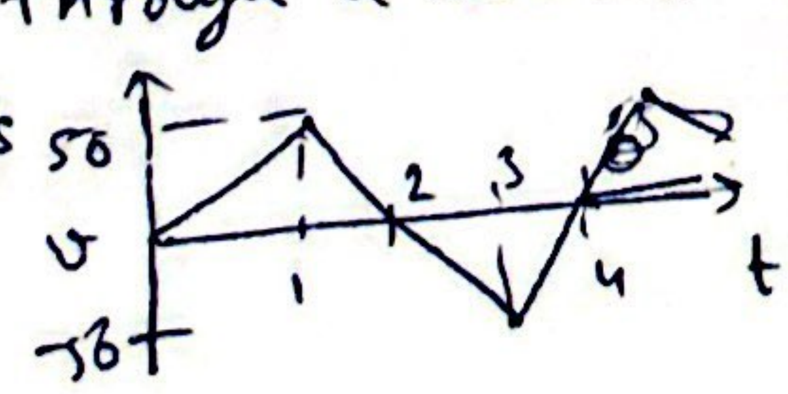


if switch closes at $t = 0$ & cap is initially relaxed, initial condition $i(0^+)$, $v_c(0^+)$, $v_c'(0^+)$

$v_c(0^+) = 0 \Rightarrow i(0^+) = \frac{V}{R}$

$\frac{dv_c}{dt}(0^+) = i/C = \frac{V}{RC}$

Qu: determine the current as fn. of time through a 200 μF whose voltage is



$$v = \begin{cases} 50t & 0<t<1 \\ 100-50t & 1<t<3 \\ -200+50t & 3<t<4 \\ 0 & \text{otherwise} \end{cases}$$

$$i = C\,dv/dt = \begin{cases} 10\text{ mA} & 0<t<1 \\ -10\text{ mA} & 1<t<3 \\ 10\text{ mA} & 3<t<4 \\ 0\text{ mA} & \end{cases}$$

linearity : homogeneity & Additivity

1.) homogeneity property

if $x(t) \longrightarrow y(t)$

then $\alpha x(t) \longrightarrow \alpha y(t)$

2.) Additive property

if $x_1(t) \to y_1(t)$ and

$x_2(t) \to y_2(t)$

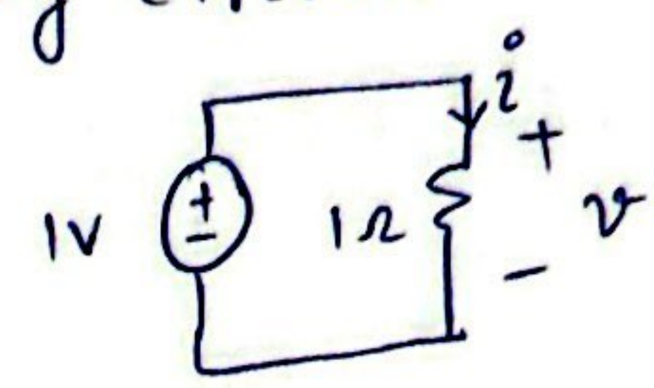
then $x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t)$

Check if R, L, C are linear

KCL - KVL

analyzing circuits

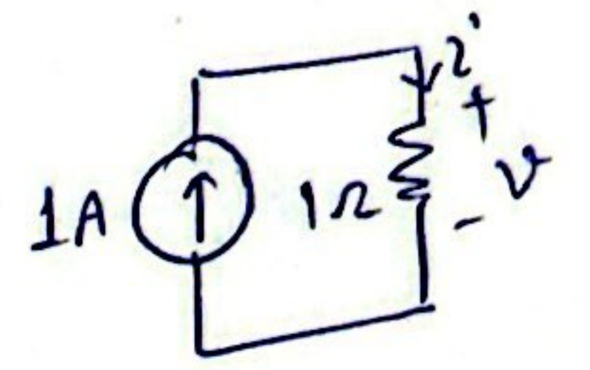
Ex.



$v = 1V$

$i = \frac{1}{1} = 1A$

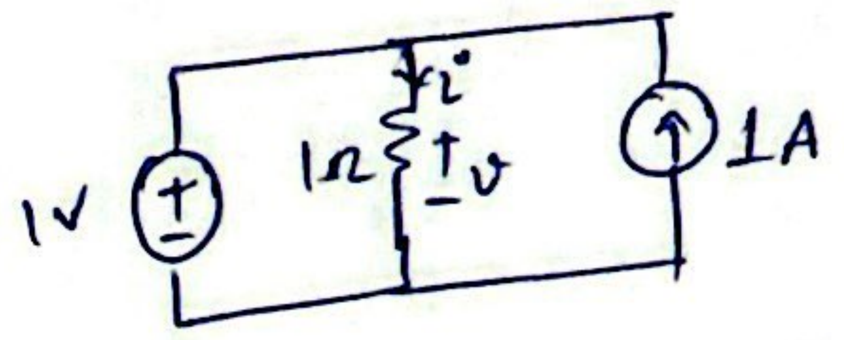
Ex.



$i = 1A$

$v = i \times R = 1V$

Qu:



$i = ?$

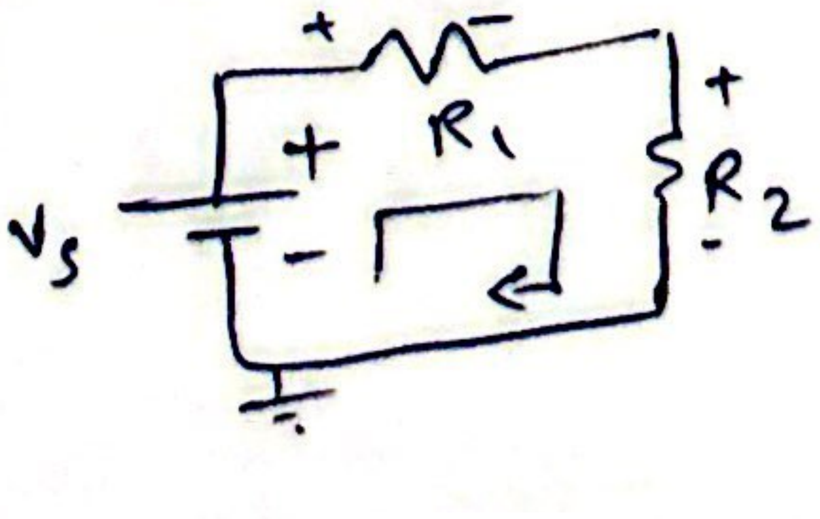
A) 1 A $\quad$ B) 2 A $\quad$ C) 0A $\quad$ D) can not determine

⇒ since voltage generated by current source is same as voltage source, voltage source is not providing any current.

⇒ if current source is 2 A, then 1 A current flows into the voltage source.

More complex ckts can be analyzed using KVL & KCL

KVL: Algebraic sum of all voltages around a closed path is zero



$-V_s + V_{R_1} + V_{R_2} = 0$

$-V_s + iR_1 + iR_2 = 0$